ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success



PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Error in Measurements & Instruments

English Medium

EXERCISE (O)

A. Significant digits

1. A wire has a mass 0.3 ± 0.003 g, radius 0.5 ± 0.005 mm and length 6 ± 0.06 cm. The maximum percentage error in the measurement of its density is :- **[JEE 2004]**

(A) 1 (B) 2 (C) 3 (D) 4

ER0018

2. The edge of a cube is a = 1.2×10^{-2} m. Then its volume will be recorded as : **[JEE 2003]**

(A) 1.7×10^{-6} m^3 (B) 1.70×10^{-6} m^3 (C) 1.70×10^{-7} m^3 (D) 1.78×10^{-6} m^3

ER0019

3. A vernier callipers having 1 main scale division = 0.1 cm is designed to have a least count of 0.02 cm. If n be the number of divisions on vernier scale and m be the length of vernier scale, then :-

(A) n = 10, m = 0.5 cm (B) n = 9, m = 0.4 cm

(C) n = 10, m = 0.8 cm (D) n = 10, m = 0.2 cm

ER0020

4. Accuracy and precision are ____(i)____ and these are respectively linked with ____(ii)____ & ____(iii)____ . Fill the blanks above in correct order.

(A) (i) same, (ii) systematic error, (iii) random error

(B) (i) different, (ii) systematic error (iii) random error

(C) (i) same, (ii) random error, (iii) systematic error

(D) (i) different, (ii) random error, (iii) systematic error

ER0031

B. Vernier callipers & Screw gauge

5. On the basis of detail given about two measuring instruments, select the correct statement.

(i) Vernier callipers having main scale division = 0.05 cm and Vernier scale division = $\frac{2.45}{50}$ cm.

(ii) Screw gaugae having pitch 0.5 mm and its circular scale division measures 0.01 mm.

(A) Both the instruments have same least count.

(B) Least count of screw gauge is lesser than that of vernier callipers.

(C) Both the instruments have same least count but screw gauge is more precise.

(D) Both the instruments have different least count and screw gauge is more precise.

ER0027

6. In which of the following instruments used in the lab there may exist back lash error.

(i) Screw gauge (ii) Spherometer (iii) Searle's apparatus (iv) Vernier callipers

(A) (i) & (ii) only (B) (i), (ii) & (iii) only (C) (i) only (D) all four

ER0029

7. The vernier of a circular scale is divided in to 30 divisions, which coincides with 29 main scale divisions. If each main scale division is $(1/2)^o$, the least count of the instrument is

(A) 0.1' (B) 1' (C) 10' (D) 30'

ER0032

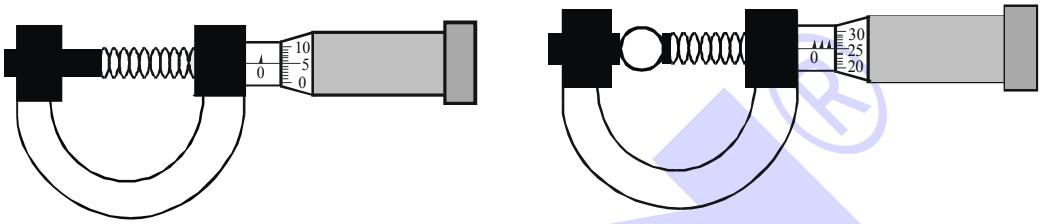
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8. When the gap is closed without placing any object in the screw gauge whose least count is 0.005 mm, the 5th division on its circular scale coincides with the reference line on main scale, and when a small sphere is placed reading on main scale advances by 4 divisions, whereas circular scale reading advances to five times to the corresponding reading when no object was placed. There are 200 divisions on the circular scale. The radius of the sphere is

(A) 4.10 mm (B) 4.05 mm (C) 2.10 mm (D) 2.05 mm

ER0033

9. The circular divisions of shown screw gauge are 50. It moves 0.5 mm on main scale in one rotation. The diameter of the ball is **[JEE 2006]**



(A) 2.25 mm (B) 2.20 mm (C) 1.20 mm (D) 1.25 mm

ER0073

10. In ordinary Vernier calipers, 10th division of the Vernier scale coincides with 9th division of the main scale. In a specially designed Vernier calipers the Vernier scale is so constructed that 10th division on it coincides with 11th division on the main scale. Each division on the main scale equals to 1 mm. The calipers have a zero error as shown in the figure-I. When the Vernier caliper is used to measure a length, the concerned portion of its scale is shown in figure-II.

Figure-I

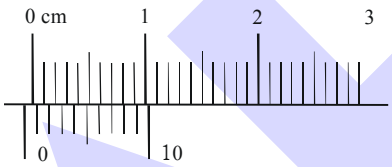
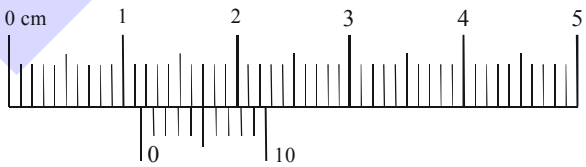


Figure-II



(A) Zero error in the calipers has magnitude 0.7 mm.
(B) The length being measured is 1.08 cm.
(C) The length being measured is 1.22 cm.
(D) Zero error in the calipers has magnitude 0.3 mm

ER0041

11. A spectrometer gives the following reading when used to measure the angle of a prism. Main scale reading : 58.5 degree
Vernier scale reading : 09 divisions
Given that 1 division on main scale corresponds to 0.5 degree. Total divisions on the vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data :

[AIEEE - 2012]

(A) 59 degree (B) 58.59 degree (C) 58.77 degree (D) 58.65 degree

ER0049

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C. Error analysis in experiments

12. In the Searle's experiment, after every step of loading, why should we wait for two minutes before taking the readings? (More than one correct.)

(A) So that the wire can have its desired change in length.

(B) So that the wire can attain room temperature.

(C) So that vertical oscillations can get subsided.

(D) So that the wire has no change in its radius.

ER0021

13. In a meter bridge set up, which of the following should be the properties of the one meter long wire?

(A) High resistivity and low temperature coefficient

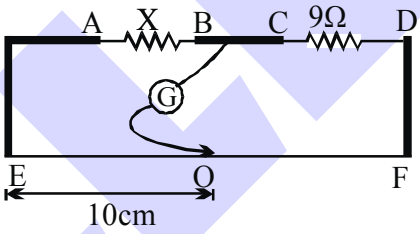
(B) Low resistivity and low temperature coefficient

(C) Low resistivity and high temperature coefficient

(D) High resistivity and high temperature coefficient

ER0022

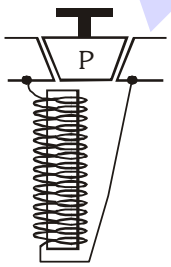
14. Consider the MB shown in the diagram, let the resistance X have temperature coefficient α_1 and the resistance from the RB have the temperature coefficient α_2. Let the reading of the meter scale be 10cm from the LHS. If the temperature of the two resistance increase by small temperature ΔT then what is the shift in the position of the null point? Neglect all the other changes in the bridge due to temperature rise.

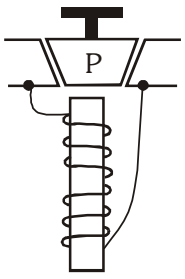


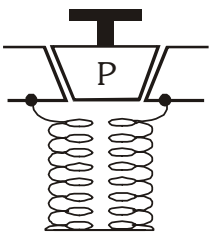
(A) $9(\alpha_1 - \alpha_2)\Delta T$ (B) $9(\alpha_1 + \alpha_2)\Delta T$ (C) $\frac{1}{9}(\alpha_1 + \alpha_2)\Delta T$ (D) $\frac{1}{9}(\alpha_1 - \alpha_2)\Delta T$

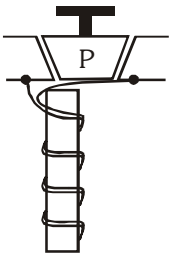
ER0023

15. Identify which of the following diagrams represent the internal construction of the coils wound in a resistance box or PO box?

(A) 

(B) 

(C) 

(D) 

ER0024

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16. In a meter bridge experiment, we try to obtain the null point at the middle. This
(A) reduces systematic error as well as random error.
(B) reduces systematic error but not the random error.
(C) reduces random error but not the systematic error
(D) reduces neither systematic error nor the random error

ER0025

17. In a Searle's experiment for determination of Young's Modulus, when a load of 50 kg is added to a 3 meter long wire micrometer screw having pitch 1 mm needs to be given a quarter turn in order to restore the horizontal position of spirit level. Young's modulus of the wire if its cross sectional area is 10^{-5} m^2 is
(A) $6 \times 10^{11} N/m^2$ (B) 1.5×10^{11} N/m^2 (C) 3×10^{11} N/m^2 (D) None

ER0026

18. A student obtained following observations in an experiment of meter bridge to find unknown resistance of given wire :

S.No.	**R**	ℓ	**100 – ℓ**	$S = \left(\frac{100-\ell}{\ell}\right) R$
1	2Ω	43	57	2.65
2	3Ω	52	48	2.77
3	4Ω	58	42	2.89
4	5Ω	69	31	2.25

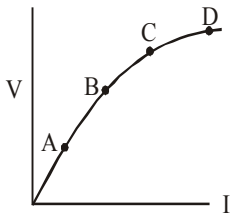
The most accurate value of unknown resistance will be
(A) 2.65 Ω (B) 2.77 Ω (C) 2.89 Ω (D) 2.25 Ω

ER0028

19. In Searle's apparatus, when experimental wire is loaded and unloaded, the air bubble in spirit level gets shifted.
(A) towards reference wire while loading & towards experimental wire while unloading
(B) towards experimental wire while loading & towards reference wire while unloading
(C) towards experimental wire, both the times, during loading & unloading
(D) towards reference wire, both the times during loading & unloading

ER0030

20. Variation of current passing through a conductor as the voltage supplied across its ends as varied is shown in the adjoining diagram. If the resistance (R) is determined at the points A, B, C and D we will find that -

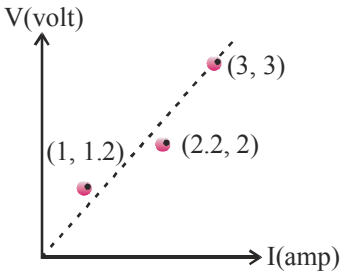


(A) $R_C = R_D$ (B) $R_B > R_A$ (C) $R_C > R_B$ (D) $R_A > R_C$

ER0072

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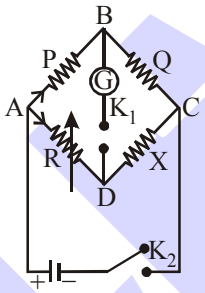
21. In the measurement of resistance of a wire using Ohm's law, the plot between V and I is drawn as shown. The resistance of the wire is -



(A) 0.833 Ω (B) 0.9 Ω (C) 1 Ω (D) None of these

ER0034

22. In Wheatstone bridge experiment as shown in figure –



(A) Key K_1 should be pressed first and then K_2
(B) Key K_2 should be pressed first and then K_1
(C) any key can be pressed in any order
(D) both keys should be pressed simultaneously.

ER0035

23. In a metre bridge experiment null point is obtained at 20 cm from one end of the wire when resistance X is balanced against another resistance Y. If X < Y, then where will be the new position of the null point from the same end, if one decide to balance a resistance of 4X against Y -

(A) 50 cm (B) 80 cm
(C) 40 cm (D) 70 cm

ER0036

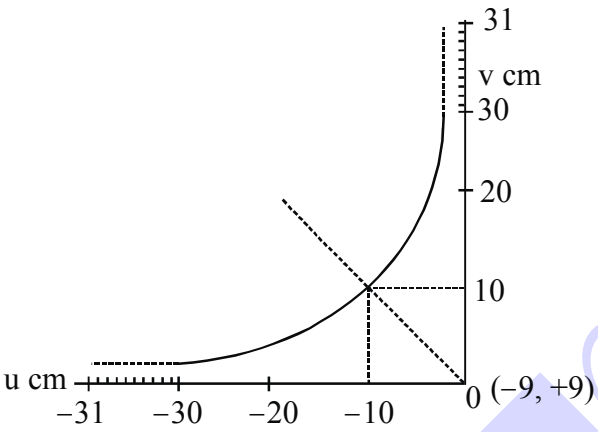
24. In a resonance column method, resonance occurs at two successive level of $\ell_1 = 30.7$ cm and $\ell_2 = 63.2$ cm using a tuning fork of f = 512 Hz. What is the maximum error in measuring speed of sound using relations $v = f\lambda$ & $\lambda = 2(\ell_2 - \ell_1)$ **[JEE 2005]**

(A) 256 cm/sec (B) 92 cm/sec
(C) 128 cm/sec (D) 102.4 cm/sec

ER0037

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25. Graph of position of image vs position of point object from a convex lens is shown. Then, focal length of the lens is **[JEE 2006]**



(A) 0.50 ± 0.05 cm (B) 0.50 ± 0.10 cm (C) 5.00 ± 0.05 cm (D) 5.00 ± 0.10 cm

ER0038

26. A student performs an experiment for determination of $g\left(=\frac{4\pi^2\ell}{T^2}\right)$ $\ell \approx 1$m and he commits an error of Δl. For the experiment takes the time of n oscillations with the stop watch of least count ΔT and he commits a human error of 0.1sec. For which of the following data, the measurement of g will be most accurate? **[JEE 2006]**

	$\Delta\ell$	ΔT	n	Amplitude of oscillation
(A)	5 mm	0.2 sec	10	5 mm
(B)	5 mm	0.2 sec	20	5 mm
(C)	5 mm	0.1 sec	20	1 mm
(D)	1 mm	0.1 sec	50	1 mm

ER0039

27. In an experiment to determine the focal length (f) of a concave mirror by the u-v method, a student places the object pin A on the principal axis at a distance x from the pole P. The student looks at the pin and its inverted image from a distance keeping his/her eye in line with PA. When the student shifts his/her eye towards left, the image appears to the right of the object pin. Then, **[JEE 2007]**

(A) $x < f$ (B) $f < x < 2f$ (C) $x = 2f$ (D) $x > 2f$

ER0040

28. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of ± 0.05 mm at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of ± 0.01 mm. Take g = 9.8 m/s^2 (exact). The Young's modulus obtained from the reading is **[JEE 2007]**

(A) $(2.0 \pm 0.3) \times 10^{11}$ N/m^2 (B) $(2.0 \pm 0.2) \times 10^{11}$ N/m^2

(C) $(2.0 \pm 0.1) \times 10^{11}$ N/m^2 (D) $(2.0 \pm 0.05) \times 10^{11}$ N/m^2

ER0042

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29. An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are meausred by **[AIEEE - 2008]**

(A) a vernier scale provided on the microscope

(B) a stanard laboratory scale

(C) a meter scale provided on the microscope

(D) a screw gauge provided on the microscope

ER0043

30. In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v, from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be :- **[AIEEE - 2009]**

(A) (f, f) (B) $(4f, 4f)$ (C) $(2f, 2f)$ (D) $\left(\frac{f}{2}, \frac{f}{2}\right)$

ER0046

31. Students I, II and III perform an experiment for measuring the acceleration due to gravity (g) using pendulum. They use different lengths of the pendulum and/or record time for different number of oscillations. The observations are shown in the table. **[JEE 2008]**

Least count for length = 0.1 cm Least count for time = 0.1 s

Student	Length of the Pendulum (cm)	Number of oscilltions (n)	Total time for (n) oscillations (s)	Time period (s)
I	64.0	8	128.0	16.0
II	64.0	4	64.0	16.0
III	20.0	4	36.0	9.0

If E_I, E_{II} and E_{III} are the percentage error in g, i.e., $\left(\frac{\Delta g}{g} \times 100\right)$ for students I, II and III, respectively,

(A) $E_I = 0$ (B) E_I is minimum (C) $E_I = E_{II}$ (D) E_{II} is maximum

ER0074

32. A student uses a simple pendulum of exactly 1m length to determine g, the acceleration due to gravity. He uses a stopwatch with the least count of 1 sec for this and records 40 seconds for 20 oscillations. For this observation, which of the following statement(s) is (are) true? **[JEE 2010]**

(A) Error ΔT in measuring T, the time period is 0.05 seconds

(B) Error ΔT in measuring T, the time period is 1 second

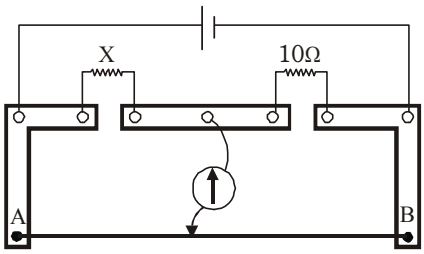
(C) Percentage error in the determination of g is 5%

(D) Percentage error in the determination of g is 2.5%

ER0075

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33. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10 ohm resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B. The determined value of 'X' is **[JEE 2011]**



(A) 10.2 ohm (B) 10.6 ohm (C) 10.8 ohm (D) 11.1 ohm

ER0078

34. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is **[JEE 2011]**

(A) 0.9% (B) 2.4% (C) 3.1% (D) 4.2%

ER0079

35. In an experiment to measure surface tension of completely wetting liquid, the capillary was dipped in liquid. The level of water in beaker was measured using travelling microscope to be 1.43cm. The level of water in capillary was measured to be 6.23cm. The readings for inner diameter were 0.00cm and 0.32cm respectively. Neglect the expansion of capillary and water on heating.

(A) % error in measurement of h is 0.4%

(B) % error in measurement of surface tension is 6.46%

(C) if temperature increases, level of water in capillary will be less than 6.23cm.

(D) $s = 0.38 \pm 0.04 N/m$

ER0105

36. In the determination of Young's modulus $\left(Y = \frac{4MLg}{\pi \ell d^2}\right)$ by using Searle's method, a wire of length L = 2m and diameter d = 0.5 mm is used. For a load M = 2.5 kg, an extension l = 0.25 mm in the length of the wire is observed. Quantities d and l are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm. The number of divisions on their circular scale is 100. The contributions to the maximum probable error of the Y measurement **[JEE 2012]**

(A) due to the errors in the measurements of d and l are the same.

(B) due to the error in the measurement of d is twice that due to the error in the measurement of l.

(C) due to the error in the measurement of l is twice that due to the error in the measurement of d.

(D) due to the error in the measurement of d is four times that due to the error in the measurement of l.

ER0080

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EXERCISE (S)

A. Significant digits

1. How many significant figures are given in the following quantities ?

(A) 343 g (B) 2.20 (C) 1.103 N (D) 0.4142 s

(E) 0.0145 m (F) 1.0080 V (G) 9.1×10^4 km (H) 1.124×10^{-3} V

ER0001

2. Perform the following operations:

(A) 703 + 7 + 0.66 (B) 2.21×0.3 (C) 12.4×84 (D) 14.28/0.714

ER0002

3. Solve with due regard to significant digits

(i) $\sqrt{6.5 - 6.32}$ (ii) $\dfrac{2.91 \times 0.3842}{0.080}$

ER0003

4. 5.74 gm of a substance occupies a volume of 1.2 cm^3. Calculate its density with due regard for significant figures.

ER0012

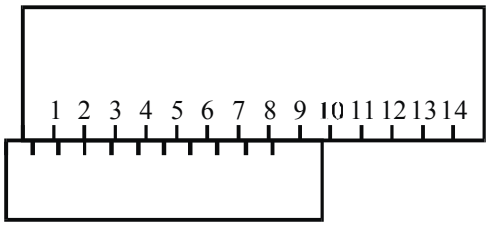
B. Vernier callipers & Screw gauge

5. The main scale of a vernier calipers reads in millimeter and its vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. When the two jaws of the instrument touch each other the seventh division of the vernier scale coincide with a scale division and the zero of the vernier lies to the right of the zero of main scale. Furthermore, when a cylinder is tightly placed along its length between the two jaws, the zero of the vernier scale lies slightly to the left of 3.2 cm; and the fourth vernier division coincides with a scale division. Calculate the measured length of the cylinder.

ER0004

6. The VC shown in the diagram has zero error in it (as you can see). It is given that 9 msd = 10 vsd.

(i) What is the magnitude of the zero error? (1msd = 1 mm)

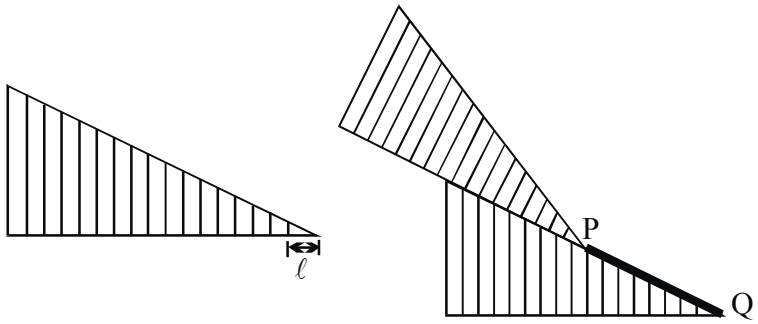


(ii) The observed reading of the length of a rod measured by this VC comes out to be 5.4 mm. If the vernier had been error free then reading of main scale would be ___ and the coinciding division of vernier scale would be _____.

ER0005

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7. Consider a home made vernier scale as shown in the figure.



In this diagram, we are interested in measuring the length of the line PQ. If both the inclines are identical and their angles are equal to θ then what is the least count of the instrument.

ER0065

8. The pitch of a screw gauge is 0.5 mm and there are 50 divisions on the circular scale. In measuring the thickness of a metal plate, there are five divisions on the pitch scale (or main scale) and thirty fourth division coincides with the reference line. Calculate the thickness of the metal plate.

ER0006

9. The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line zero of the main scale is not visible. When a glass plate is placed between the studs, the main scale reads three divisions and the circular scale reads 26 divisions. Calculate the thickness of the plate.

ER0007

10. The pitch of a screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a wire, the linear scale reads 1 mm and 47th division on the circular scale coincides with the reference line. The length of the wire is 5.6 cm. Find the curved surface area (in cm^2) of the wire in appropriate number of significant figures. **[JEE 2004]**

ER0015

11. In a vernier calipers the main scale and the vernier scale are made up of different materials. When the room temperature increases by $\Delta T°C$, it is found the reading of the instrument remains the same. Earlier it was observed that the front edge of the wooden rod placed for measurement crossed the Nth main scale division and (N + 2) msd coincided with the 2nd vsd. Initially, 10 vsd coincided with 9 msd. If coefficient of linear expansion of the main scale is α_1 and that of the vernier scale is α_2 then what is the value of α_1/α_2? (Ignore the expansion of the rod on heating)

ER0069

12. In a vernier callipers, n divisions of its main scale match with (n + 1) divisions on its vernier scale. Each division of the main scale is a units. Using the vernier principle, calculate its least count.

[JEE 2003]

ER0070

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2-EM & INS. Semiconductor-PCS\Eng\3.Error Measu & Instr\0.1_Th + Ex.p65

13. The side of a cube is measured by vernier callipers (10 divisions of a vernier scale coincide with 9 divisions of main scale, where 1 division of main scale is 1 mm). The main scale reads 10 mm and first division of vernier scale coincides with the main scale. Mass of the cube is 2.736 g. Find the density of the cube in appropriate significant figures. **[JEE 2005]**

ER0017

C. Error analysis in experiments

14. In a given optical bench, a needle of length 10 cm is used to estimate bench error. The object needle, image needle & lens holder have their reading as shown.

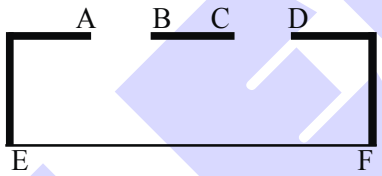
$x_0 = 1.1$ cm $\quad x_I = 21.0$ cm $\quad x_L = 10.9$ cm

Estimate the bench errors which are present in image needle holder and object needle holder. Also find the focal length of the convex lens when.

$x_0 = 0.6$ cm $\quad x_I = 22.5$ cm $\quad x_L = 11.4$ cm

ER0066

15. Make the appropriate connections in the meter bridge set up shown. Resistance box is connected between ___. Unknown resistance is connected between___. Battery is connected between _____.



ER0008

16. A body travels uniformly a distance of (13.8 ± 0.2)m in time (4.0 ± 0.3) sec. Calculate its velocity.

ER0009

17. Consider S = x cos (θ) for x = (2.0 ± 0.2) cm, θ = 53 ± 2 °. Find S.

ER0010

18. Two resistance R_1 and R_2 are connected in (i) series and (ii) parallel. What is the equivalent resistance with limit of possible percentage error in each case of $R_1 = 5.0 \pm 0.2\ \Omega$ and $R_2 = 10.0 \pm 0.1\ \Omega$.

ER0011

19. The time period of oscillation of a simple pendulum is given by $T = 2\pi\sqrt{\ell / g}$. The length of the pendulum is measured as $\ell = 10.0 \pm 0.1$cm and the time period as $T = 0.50 \pm 0.02$ s. Determine percentage error in the value of g.

ER0013

20. In a Searle's experiment, the diameter of the wire as measured by a screw gauge of least count 0.001 cm is 0.050 cm. The length, measured by a scale of least count 0.1 cm, is 110.0 cm. When a weight of 50 N is suspended from the wire, the extension is measured to be 0.125 cm by a micrometer of least count 0.001 cm. Find the maximum error in the measurement of Young's modulus of the material of the wire from these data. **[JEE 2004]**

ER0014

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21. A physical quantity P is related to four observables A, B, C and D as $P = 4\pi^2 A^3 B^2 / (\sqrt{C}\, D)$

The percentage error of the measurement in A, B, C and D are 1%, 3% and 2%, 4% respectively. Determine the percentage error & absolute error in the quantity P. Value of P is calculated 3.763.

ER0067

22. A glass prism of angle A = 60° gives minimum angle of deviation θ = 30° with the max. error of 1^0 when a beam of parallel light passed through the prism during an experiment. Find the permissible error in the measurement of refractive index μ of the material of the prism.

ER0068

23. The period of oscillation of a simple pendulum in an experiment is recorded as 2.63 sec, 2.56 sec, 2.42 sec, 2.71 sec and 2.80 sec respectively. Find the time period, the absolute error in each observation, average absolute error and the percentage error.

ER0016

24. A travelling microscope is used to measure the refractive index of a glass slab. The microscope is focused on a spot on the table and the reading of microscope is 10.08cm. Now the glass slab is put on the spot & microscope is raised to focus on the apparent position of spot. The reading is 12.48cm. Now the microscope is raised further to focus on the top of the slab. The reading is 14.88cm. If the least count of scale is 0.01cm, what is the % error in refractive index?

ER0104

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EXERCISE-JM

1. The current voltage relation of diode is given by $I = (e^{1000V/T} - 1)$ mA, where the applied voltage V is in volts and the temperature T is in degree Kelvin. If a student makes an error measuring ± 0.01V while measuring the current of 5 mA at 300 K, what will be error in the value of current in mA ?

[JEE-Main 2014]

(1) 0.5 mA (2) 0.05 mA (3) 0.2 mA (4) 0.02 mA

ER0051

2. A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it ? **[JEE-Main 2014]**

(1) A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.

(2) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

(3) A meter scale

(4) A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.

ER0052

3. The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90s using a wrist watch of 1s resolution. The accuracy in the determination of g is : **[JEE-Main 2015]**

(1) 1% (2) 5% (3) 2% (4) 3%

ER0053

4. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement , it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line ? **[JEE-Main 2016]**

(1) 0.50 mm (2) 0.75 mm (3) 0.80 mm (4) 0.70 mm

ER0054

5. A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be :- **[JEE-Main 2016]**

(1) 92 ± 3 s (2) 92 ± 2 s (3) 92 ± 5.0 s (4) 92 ± 1.8 s

ER0055

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6. To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ. If a shunt of resistance S is needed to get half deflection then G, R and S are related by the equation :

[JEE-Mains (Online) 2016]

(1) 2S (R+G) = RG (2) S (R+G) = RG (3) 2G = S (4) 2S = G

ER0056

7. The following observations were taken for determining surface tensiton T of water by capillary method:
Diameter of capilary, D = 1.25×10^{-2} m
rise of water, h = 1.45×10^{-2} m

Using g = 9.80 m/s^2 and the simplified relation T = $\frac{rhg}{2} \times 10^3$ N/m, the possible error in surface tension is closest to : **[JEE-Main 2017]**

(1) 2.4% (2) 10% (3) 0.15% (4) 1.5%

ER0057

8. The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is :- **[JEE-Main 2018]**

(1) 3.5 % (2) 4.5 % (3) 6 % (4) 2.5 %

ER0058

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

9. The pitch and the number of divisions, on the circular scale, for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 divisions below the mean line.
The readings of the main scale and the circular scale, for a thin sheet, are 5.5 mm and 48 respectively, the thickness of this sheet is : **[JEE Main-2019_Jan.]**

(1) 5.755 m (2) 5.725 mm (3) 5.740 m (4) 5.950 mm

ER0093

10. The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figures ? **[JEE Main-2019_Jan.]**

(1) 4260 ± 80 cm^3 (2) 4300 ± 80 cm^3
(3) 4264.4 ± 81.0 cm^3 (4) 4264 ± 81 cm^3

ER0094

11. The least count of the main scale of a screw gauge is 1 mm. The minimum number of divisions on its circular scale required to measure 5μm diameter of wire is : **[JEE Main-2019_Jan.]**

(1) 50 (2) 100 (3) 200 (4) 500

ER0095

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2, EM & INS. Semiconductor-PCS\Eng\3.Error Measu & Instr\01_Th + Ex.p65

12. In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to :-

[JEE Main-2019_April]

(1) 0.7% (2) 0.2% (3) 3.5% (4) 6.8%

ER0096

13. The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division) **[JEE Main-2020_Sep.]**

(1) 3.21 cm (2) 2.99 cm (3) 3.2 cm (4) 3.07 cm

ER0098

14. Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as : **[JEE Main-2020_Sep.]**

(1) 2.123 cm (2) 2.125 cm (3) 2.121 cm (4) 2.124 cm

ER0099

15. A physical quantity z depends on four observables a, b, c and d, as $z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c}\, d^3}$. The percentage of error in the measurement of a, b, c and d 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is: **[JEE Main-2020_Sep.]**

(1) 12.25% (2) 14.5% (3) 16.5% (4) 13.5%

ER0100

16. The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is . **[JEE Main-2020_Sep.]**

ER0101

17. A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm; 5.65 mm. The average of these four readings is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as: **[JEE Main-2020_Sep.]**

(1) (5.5375 ± 0.0739) mm (2) (5.538 ± 0.074) mm

(3) (5.54 ± 0.07) mm (4) (5.5375 ± 0.0740) mm

ER0102

node06\B0BC-BD\Kota\IEE(Advanced)\Enthuse\Phy\Module\5-Mod Phy-1, 2-EM & INS.-Semiconductor-PCS\Eng\3-Error Measu & Instr\01_Th + Ex.p65

18. In order to determine the Young's Modulus of a wire of radius 0.2 cm (measured using a scale of least count = 0.001 cm) and length 1m (measured using a scale of least count = 1 mm), a weight of mass 1kg (measured using a scale of least count = 1g) was hanged to get the elongation of 0.5 cm (measured using a scale of least count 0.001 cm). What will be the fractional error in the value of Young's Modulus determined by this experiment ? **[JEE Main-2021_March]**

(1) 0.14% (2) 0.9% (3) 9% (4) 1.4%

ER0106

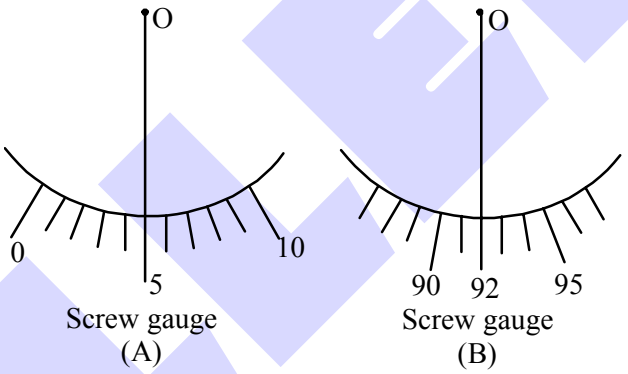
19. In the experiment of Ohm's law, a potential difference of 5.0 V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is :-

[JEE Main-2021_March]

(1) 3.9 (2) 8.4 (3) 7.5 (4) 3.0

ER0107

20. Student A and Student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____. [Figure shows position of reference 'O' when jaws of screw gauge are closed] Given pitch = 0.1 cm. **[JEE Main-2021_July]**



ER0108

21. A student determined Young's Modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$. The value of g is taken to be 9.8 m/s^2, without any significant error, his observation are as following.

Physical Quantity	Least count of the Equipment used for measurement	Observed value
Mass (M)	1 g	2 kg
Length of bar (L)	1 mm	1 m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

Then the fractional error in the measurement of Y is : **[JEE Main-2021_Aug]**

(1) 0.0083 (2) 0.0155 (3) 0.155 (4) 0.083

ER0109

node06\B09BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2, EM & INS. Semiconductor-PCS\Eng\3.Error Measu & Instr\01_Th + Ex.p65

22. A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is _____ $\times 10^{-6}$ m. **[JEE Main-2022_June]**

ER0111

23. A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22mm, 1.23 mm, 1.19 mm and 1.20 mm. The percentage error is $\frac{x}{121}\%$. The value of x is ___

[JEE Main-2022_June]

ER0112

24. A screw gauge of pitch 0.5mm is used to measure the diameter of uniform wire of length 6.8cm, the main scale reading is 1.5 mm and circular scale reading is 7. The calculated curved surface area of wire to appropriate significant figures is : [Screw gauge has 50 divisions on the circular scale]

[JEE Main-2022_July]

(A) 6.8 cm^2 (B) 3.4 cm^2 (C) 3.9 cm^2 (D) 2.4 cm^2

ER0113

25. In an experiment of determine the Young's modulus of wire of a length exactly 1m, the extension in the length of the wire is measured as 0.4mm with an uncertainty of $\pm$ 0.02 mm when a load of 1kg is applied. The diameter of the wire is measured as 0.4mm with an uncertainty of $\pm$ 0.01 mm. The error in the measurement of Young's modulus (ΔY) is found to be $x \times 10^{10}$ Nm^{-2}. The value of x is _____
[Take : $g = 10$ m/s^2] **[JEE Main-2022_July]**

ER0114

26. A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of :

[JEE Main-2022_July]

(A) 15% (B) 25% (C) 75% (D) 5%

ER0115

27. The one division of main scale of vernier callipers reads 1 mm and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between 4.1 cm and 4.2 cm and 6th Vernier division coincides with a main scale division. The diameter of the bob will be_____ 10^{-2} cm

[JEE Main-2022_July]

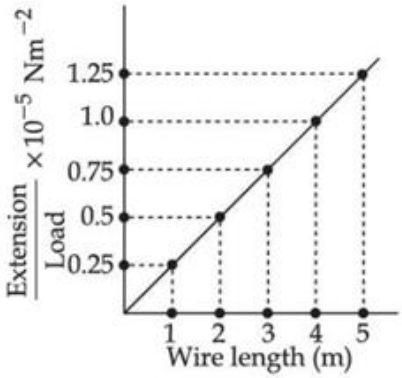
ER0116

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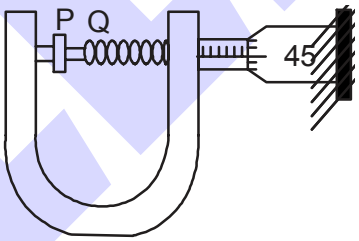
28. In an experiment to determine the Young's modulus, steel wires of five different lengths (1, 2, 3, 4 and 5 m) but of same cross section (2 mm^2) were taken and curves between extension and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}$ Nm^{-2}, then the value of x is _____ .

[JEE Main-2022_July]

ER0117



29. In an experiment to find out the diameter of wire using screw gauge, the following observation were noted :



(a) Screw moves 0.5 mm on main scale in one complete rotation

(b) Total divisions on circular scale = 50

(c) Main scale reading is 2.5 mm

(d) 45th division of circular scale is in the pitch line

(e) Instrument has 0.03 mm negative error

Then the diameter of wire is : **[JEE Main-2022_July]**

(A) 2.92 mm (B) 2.54 mm (C) 2.98 mm (D) 3.45 mm

ER0118

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EXERCISE - JA

1. The diameter of a cylinder is measured using a Vernier callipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale excatly coincides with one of the main scale divisions. The diameter of the cylinder is :- **[JEE-Advance 2013]**

(A) 5.112 cm (B) 5.124 cm (C) 5.136 cm (D) 5.148 cm

ER0081

2. Using the expression $2d \sin\theta = \lambda$, one calculates the values of d by measuring the corresponding angles θ in the range 0 to 90°. The wavelength λ is exactly known and the error in θ is constant for all values of θ. As θ increases from 0° :- **[JEE-Advance 2013]**

(A) the absolute error in d remains constant

(B) the absolute error in d increases

(C) the fractional error in d remains constant

(D) the fractional error in d decreases

ER0082

3. During Searle's experiment, zero of the Vernier scale lies between 3.20×10^{-2} m and 3.25×10^{-2}m of the main scale. The 20th division of the Vernier scale exactly coincides with one of the main scale divisions. When an additional load of 2 kg is applied to the wire, the zero of the Vernier scale still lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale but now the 45th division of Vernier scale coincides with one of the main scale divisions. The length of the thin metallic wire is 2m and its cross-sectional area is 8×10^{-7} m^2. The least count of the Vernier scale is 1.0×10^{-5} m. The maximum percentage error in the Young's modulus of the wire is. **[JEE-Advance 2014]**

ER0083

4. Consider a Vernier callipers in which each 1 cm on the main scale is divided into 8 equal divisions and a screw gauge with 100 divisions on its circular scale. In the Vernier callipers, 5 divisions of the Vernier scale coincide with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Then : **[JEE-Advance 2015]**

(A) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 mm.

(B) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 mm.

(C) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 mm.

(D) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 mm.

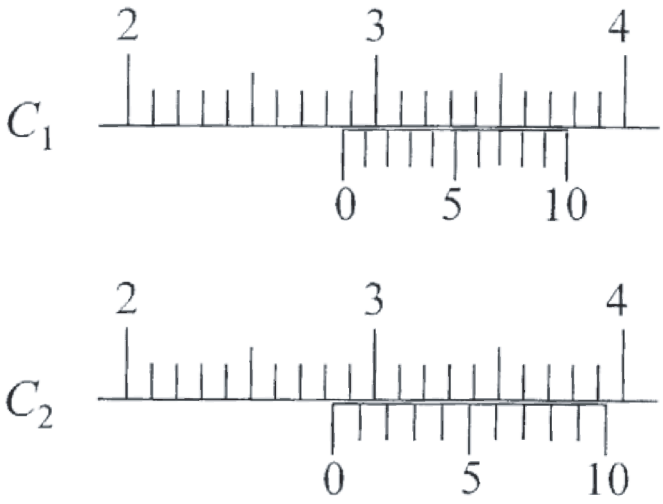
ER0084

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5. The energy of a system as a function of time t is given as $E(t) = A^2\exp(-\alpha t)$, where $\alpha = 0.2s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of E(t) at t = 5 s is. **[JEE-Advance 2015]**

ER0085

6. There are two vernier calipers both of which have 1 cm divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers (C_1) has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure. The measured values (in cm) by calipers C_1 and C_2 respectively, are **[JEE-Advance 2016]**



(A) 2.87 and 2.86 (B) 2.87 and 2.87 (C) 2.87 and 2.83 (D) 2.85 and 2.82

ER0086

7. In an experiment to determine the acceleration due to gravity g, the formula used for the time period of a periodic motion is $T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are measured to be (60 ± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52 s, 0.56 s, 0.57s, 0.54 s and 0.59 s. The least count of the watch used for the measurement of time period is 0.01 s. Which of the following statement(s) is(are) true ? **[JEE-Advance 2016]**

(A) The error in the measurement of r is 10%

(B) The error in the measurement of T is 3.57 %

(C) The error in the measurement of T is 2%

(D) The error in the determined value of g is 11%

ER0087

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\5-Mod Phy-1, 2, EM & INS. Semiconductor-PCS\Eng\3.Error Measu & Instr\01_Th + Ex.p65

8. A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the bottom of the well. The error in his measurement of time is $\delta T = 0.01$ second and he measures the depth of the well to be $L = 20$ meters. Take the acceleration due to gravity $g = 10\ ms^{-2}$ and the velocity of sound is $300\ ms^{-1}$. Then the fractional error in the measurement, $\delta L/L$, is closest to **[JEE-Advance 2017]**

(A) 0.2 % (B) 5 % (C) 3 % (D) 1 %

ER0088

PARAGRAPH "A"

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x, y and z are Δx, Δy and Δz, respectively, then **[JEE-Advance 2018]**

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y / y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1, \frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

(There are two questions based on Paragraph "A", the question given below is one of them)

9. Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a. If the error in the measurement of a is $\Delta a (\Delta a/a << 1)$, then what is the error Δr in determining r ?

(A) $\frac{\Delta a}{(1+a)^2}$ (B) $\frac{2\Delta a}{(1+a)^2}$ (C) $\frac{2\Delta a}{(1-a^2)}$ (D) $\frac{2a\Delta a}{(1-a^2)}$

ER0089

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PARAGRAPH "A"

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation z = x/y. If the errors in x, y and z are Δx, Δy and Δz, respectively, then

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y / y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1$, $\frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

(There are two questions based on Paragraph "A", the question given below is one of them)

10. In an experiment the initial number of radioactive nuclei is 3000. It is found that 1000 ± 40 nuclei decayed in the first 1.0 s. For $|x| << 1$, In (1 + x) = x up to first power in x. The error $\Delta\lambda$, in the determination of the decay constant λ, in s^{-1}, is :-

(A) 0.04 (B) 0.03 (C) 0.02 (D) 0.01

ER0090

11. A steel wire of diameter 0.5 mm and Young's modulus 2×10^{11} N m^{-2} carries a load of mass M. The length of the wire with the load is 1.0 m. A vernier scale with 10 divisions is attached to the end of this wire. Next to the steel wire is a reference wire to which a main scale, of least count 1.0 mm, is attached. The 10 divisions of the vernier scale correspond to 9 divisions of the main scale. Initially, the zero of vernier scale coincides with the zero of main scale. If the load on the steel wire is increased by 1.2kg, the vernier scale division which coincides with a main scale division is..... Take g = 10 ms^{-2} and π = 3.2. **[JEE-Advance 2018]**

ER0091

12. An optical bench has 1.5 m long scale having four equal divisions in each cm. While measuring the focal length of a convex lens, the lens is kept at 75 cm mark of the scale and the object pin is kept at 45 cm mark. The image of the object pin on the other side of the lens overlaps with image pin that is kept at 135 cm mark. In this experiment, the percentage error in the measurement of the focal length of the lens is_____. **[JEE-Advance 2019]**

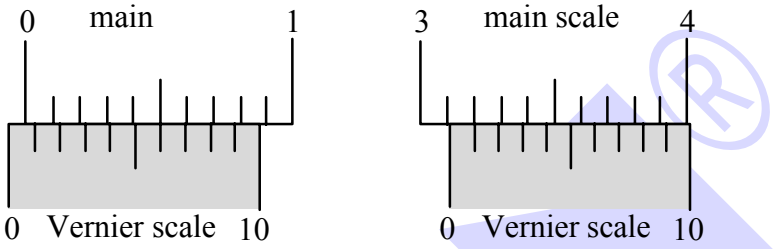
ER0092

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13. Two capacitors with capacitance values $C_1 = 2000 \pm 10$ pF and $C_2 = 3000 \pm 15$ pF are connected in series. The voltage applied across this combination is $V = 5.00 \pm 0.02$ V. The percentage error in the calculation of the energy stored in this combination of capacitors is _____. **[JEE-Advance 2020]**

ER0103

14. The smallest division on the main scale of a Vernier calipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calipers with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is **[JEE-Advance 2021]**



(A) 3.07 cm (B) 3.11 cm (C) 3.15 cm (D) 3.17 cm

ER0110

15. Area of the cross-section of a wire is measured using a screw gauge. The pitch of the main scale is 0.5mm. The circular scale has 100 divisions and for one full rotation of the circular scale, the main scale shifts by two divisions. The measured readings are listed below. **[JEE-Advance 2022]**

Measurement condition	**Main scale reading**	**Circular scale reading**
Two arms of gauge touching each other without wire	0 division	4 division
Attempt-1: With wire	4 divisions	20 divisions
Attempt-2: With wire	4 divisions	16 divisions

What are the diameter and cross-sectional area of the wire measured using the screw gauge?

(A) 2.22 ± 0.02 mm, $\pi (1.23 \pm 0.02)$ mm^2

(B) 2.22 ± 0.01 mm, $\pi (1.23 \pm 0.01)$ mm^2

(C) 2.14 ± 0.02 mm, $\pi (1.14 \pm 0.02)$ mm^2

(D) 2.14 ± 0.01 mm, $\pi (1.14 \pm 0.01)$ mm^2 **ER0119**

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ANSWER KEY

EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	C	B	A	B	B	D	C	A,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A,B,C	A	A	D	A	A	B	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	D	C	D	B	A or B	A	C
Que.	31	32	33	34	35	36				
Ans.	B	A,C	B	C	A,C,D	A				

EXERCISE # S

1. (A) 3, (B) 3, (C) 4, (D) 4, (E) 3, (F) 5, (G) 2, (H) 4

2. (A) 711, (B) 0.7, (C) 1.0×10^3, (D) 20.0 **3.** (i) 0.4, (ii) 14 **4.** 4.8 g/cm^3

5. 3.07 cm **6.** (i) x = –0.7 msd, (ii) 6, 1 **7.** L.C. = $\ell\left[\frac{1-\cos\theta}{\cos\theta}\right]$

8. 2.84 mm **9.** R_t = 3.64 mm **10.** 2.6 cm^2 (in two significant figures) **11.** 1.8 /(N+2)

12. $\frac{a}{n+1}$ **13.** 2.66 g/cm^3 **14.** 5.5 ± 0.1 cm **15.** CD, AB, EF

16. v = (3.4 ± 0.4) m/s **17.** S = (1.2 ± 0.2) cm **18.** R_8 = 15.0 Ω ± 2%, R_p = 3.3 Ω ± 3%

19. 9% **20.** $\Delta Y = 0.0489Y = 1.1 \times 10^{10}$ N/m^2 **21.** 14%, 0.53 **22.** 5π/18%

23. T = 2.62 s. 0.01 s, – 0.06 s, – 0.20 s, 0.09 s, 0.18 s, Average absolute error = 0.11 s, 4.2 %

24. 1.25

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	4	3	2	2	4	2	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	4	4	2	1050.00	3	4	1	13
Que.	21	22	23	24	25	26	27	28	29	
Ans.	2	5	150	B	2	B	412	2	C	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	4	B,C	4	C	A,B,D	D	B	C
Que.	11	12	13	14	15					
Ans.	3 [2.99, 3.01]	1.35 to 1.45	1.30	C	Dropped					

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